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Paper Code: MGI-102

M.Tech. (Geo-Informatics)
End Semester Examination 2024
I *Semester*
Fundamental of Geographic Information Systems

Time: Three Hours

MM: 100

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.

Q1. (2X10=20 Marks)

a. "Many definition have been proposed by different scientist and researchers as per their use", justify the statement by defining GIS as per Arohoff, ESRI, Clark and Smith. CO1

b. "Every technology has a certain history of evaluation", describe the history of GIS in the world and in India. CO1

c. "The acquisition of digital data by digitizing existing maps is comparatively cheaper, and requires less time compared to other methods" justify the statement by explaining Terrain Data from Existing Maps and Field Surveying Methods. CO1

Q2. (2X10=20 Marks)

a. "The data representation is in different kinds of variables, also known as scales that can be stored in a GIS", justify the statement by describing data representation with its variables. CO2

b. "Error in GIS may be categorized", justify the statement by describing all the sources of errors in GIS. CO2

c. "The data for GIS collected from different sources", justify the statement by describing atleast four sources of data collection. CO2

Q3. (2X10=20 Marks)

a. In land use / land cover classification of Satellite imagery the error matrix resulting from randomly sampled test pixels are given below. Find out the Producer's accuracy, User's accuracy, Overall accuracy & KHAT or KAPPA statistics and discuss the result. CO3

	U	W	F	A	B
U	397	0	0	3	12
W	0	226	1	2	1
F	3	3	313	1	0
A	1	0	0	126	0
B	2	1	0	0	52

b. Explain modeling network with a neat diagram and elaborate the keys characteristics. CO3

c. "There are many different data structure in use in GIS. In general data structure can be classified according to the type of data they use", justify the statement by describing all the types of data structure. CO3

Q4. (2X10=20 Marks)

a. Explain the terms Reclassification, Query, Buffering and Neighbouringhood functions. CO4

b. "GIS vector data and raster data have different approach of performing map overlays", justify the statement by describing different approaches. CO4

c. "Interpolation methods can be broadly classified" ", justify the statement by describing different methods of Interpolation. CO4

Q5.

(2X10=20 Marks)

a. Describe step by step a Flow chart of the methodology for mapping the spatial distribution of land- use conflicts CO5

b. Describe step by step a Flow chart of the steps involved in the study of Landslide Hazard Zonation CO5

c. Describe step by step a Flow chart showing the methodology adopted for groundwater quality mapping CO5